UCT641: IMAGE RECOGNITION AND PATTERN RECOGNITION

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2	0	2	3.0

Course Objectives: To provide the understanding the concepts of image processing and basic analytical methods to be used in various image processing techniques such as enhancement, segmentation, and recognition.

Introduction: Image processing systems and its applications. Basic image file formats.

Image formation: Geometric and photometric models; Digitization - sampling, quantization; Image definition and its representation, neighbourhood metrics.

Intensity transformations and spatial filtering: Enhancement, contrast stretching, histogram specification, local contrast enhancement; Smoothing, linear and order statistic filtering, sharpening, spatial convolution, Gaussian smoothing, DoG, LoG.

Segmentation: Pixel classification; Grey level thresholding, global/local thresholding; Optimum thresholding - Bayes analysis, Otsu method; Derivative based edge detection operators, edge detection/linking, Canny edge detector; Region growing, split/merge techniques, line detection, Hough transform.

Image/Object features extraction: Textural features - gray level co-occurrence matrix; Moments; Connected component analysis; Convex hull; Distance transform, medial axis transform, skeletonization/thinning, shape properties.

Registration: Mono-modal/multimodal image registration; Global/local registration; Transform and similarity measures for registration; Intensity/pixel interpolation.

Colour image processing: Fundamentals of different colour models - RGB, CMY, HSI, YCbCr, Lab; False colour; Pseudo colour; Enhancement; Segmentation.

Morphological Filtering Basics: Dilation and Erosion Operators, Top Hat Filters

Laboratory Work

To implement various image processing techniques studied during the course using OpenCV/MATLAB.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Learn the fundamentals of digital image formation and its processing along with its practical applications in various domains.
2. Understand and perform the various image enhancement techniques in spatial and frequency domain.

3. Learn and apply the concept of image segmentation.
4. Understand and implement feature extraction techniques for object detection and recognition
5. Elucidate the mathematical modelling of image registration and morphological filters.
6. Understand the various color models and comprehend the enhancement and segmentation techniques for the same.

Text Books:

1. Digital Image Processing. R. C. Gonzalez and R. E. Woods, Prentice Hall.

Reference Books:

1. Image Processing: The Fundamentals. Maria Petrou and Panagiota Bosdogianni, John Wiley & Sons, Ltd.
2. Digital Image Processing. K. R. Castleman:, Prentice Hall, Englewood Cliffs.
3. Visual Reconstruction. A. Blake and A. Zisserman, MIT Press, Cambridge.
4. Digital Pictures. A. N. Netravali and B. G. Haskell, Plenum Press.
5. Digital Images and Human Vision. A. B. Watson: MIT Press, Cambridge.